

Do Neural Networks Learn Structure-Preserving Maps?

A Case Study in Latent-to-Hilbert Embeddings

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<https://github.com/adnanphp/diffusion-hilbert>

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Abstract

We investigate whether neural networks can learn structure-preserving mappings from a compressed latent space to a Hilbert-space representation, such that inner products between latent points correspond to inner products between target points. Using a controlled setup—an 8-dimensional autoencoder bottleneck on MNIST, n -qubit product-state targets constructed via PCA-based angle encoding, and small MLPs trained with a combined mean-squared-error (MSE) and inner-product loss—we report four findings. First, the mapping is approximately linear, with a linear regression from the latent to the recovered angle space achieving $R^2 = 0.91$; the learned map’s primary direction is target-driven rather than input-driven, with cosine similarity 0.989 to the target’s principal direction and 0.002 to the input’s principal direction. Second, the learned map is genuinely rank-4: removing any singular direction degrades inner-product preservation by a factor of 2.5–5.9, despite a singular-value spectrum with two dominant and two small values. Third, the learned subspace does not coincide with the PCA basis used to construct the target; different random seeds recover the same primary direction but diverge in higher-rank structure. Fourth, in this controlled setting, kernel ridge regression with an RBF kernel can outperform a tuned MLP (IP error 0.0144 vs. 0.0197 after 24-configuration tuning), suggesting that for approximately linear structure-preserving mappings, classical kernel methods may be a simpler and more effective alternative. We discuss implications for representation alignment and for the design of structure-preserving neural architectures.

1 Introduction

Machine learning has increasingly focused on learning *representations* rather than direct input–output mappings. A recurring problem in this context is measuring whether two learned representations preserve the same underlying structure. Canonical correlation analysis [5], centered kernel alignment [4], and Procrustes-based alignment [1] are widely used to compare representations, but they typically assume a fixed target and ask whether a learned representation is close to it. In this paper, we ask a related but distinct question: given a latent vector z and a target Hilbert-space representation ψ , can a neural network learn a mapping $f(z) = \psi$ that preserves inner products between points?

This question has practical relevance in settings where geometric structure matters—representation alignment, structure-preserving embeddings, and classical-to-quantum mappings—and is theoretically motivated by the desire to understand when a neural network can recover geometric content rather than merely fit observations.

We investigate the question in a controlled setting. We use an 8-dimensional autoencoder bottleneck trained on MNIST, and construct a target Hilbert-space representation via PCA-based

angle encoding of the latent vectors. We train several models—linear regression, an MLP trained on MSE only, an MLP trained on MSE plus an inner-product loss, and kernel ridge regression—to map z to ψ . We evaluate them on inner-product preservation error (IP_err), reconstruction error (MSE), and pairwise distance preservation error (dist_err).

Our findings are as follows. First, the learned mapping is approximately linear, with a linear regression from z to the recovered angle space achieving $R^2 = 0.91$, and its primary direction is target-driven rather than input-driven. Second, the learned map requires all singular directions: rank reduction degrades IP_err by a factor of 2.5 to 5.9. Third, the learned subspace does not equal the PCA basis, and seed-to-seed variation appears in the higher-rank structure while the primary direction remains stable. Fourth, in this controlled setting, kernel ridge regression with an RBF kernel outperforms the tuned MLP, suggesting that classical kernel methods may be preferable when the underlying mapping is approximately linear.

The remainder of the paper is organized as follows. Section 2 reviews related work on representation alignment and kernel methods. Section 3 describes the experimental setup. Section 4 presents results. Section 5 discusses interpretation and implications. Section 6 outlines limitations and future work.

2 Related Work

Representation similarity. Comparing two learned representations has become a standard tool in deep learning analysis. Canonical correlation analysis (CCA) and its deep variants [5] compare representations by finding linear transformations that maximize correlation between projected views. Centered kernel alignment (CKA) [4] provides an alternative based on Hilbert–Schmidt independence, invariant to orthogonal transformations and isotropic scaling. Procrustes analysis [1] finds the orthogonal transformation that best aligns two point sets, and has been adapted to compare neural representations under the name orthogonal Procrustes.

These methods all assume a fixed target representation and ask whether a learned representation is close to it. Our setup is different: we do not assume that a target representation exists, but rather ask whether a network can learn a mapping *into* a specified target space while preserving geometric structure.

Kernel methods. Kernel ridge regression (KRR) [6] is a classic method for nonlinear regression that combines a linear model in a high-dimensional feature space with L_2 regularization. With an RBF kernel, KRR can approximate a wide class of smooth functions and typically serves as a strong baseline for low-dimensional regression tasks. Our experiments show that KRR outperforms a tuned MLP in our setting, which is consistent with the fact that our target mapping is approximately linear.

Quantum machine learning. Quantum feature spaces have been proposed as a route to enhanced expressivity in machine learning [2, 7]. A common construction maps classical data to quantum states via angle encoding, and one then applies a variational quantum circuit before measurement. Our angle encoding follows this approach; we do not simulate the variational part, but rather use the resulting product-state amplitudes as the target representation for a classical mapping.

Kernel methods vs. neural networks. The relationship between kernel methods and infinitely wide neural networks has been studied through the neural tangent kernel [3]. For finite-width

networks and small datasets, kernel methods frequently perform competitively or better. Our results add a specific example to this literature: when the target mapping is approximately linear, a kernel method outperforms a small MLP even when the MLP is trained with a custom loss designed for the task.

3 Method

3.1 Latent space

We train a small autoencoder (TinyAE) on 5,000 MNIST images with an 8-dimensional bottleneck. The encoder maps a flattened 28×28 image $x \in \mathbb{R}^{784}$ to a latent vector $z \in \mathbb{R}^8$, and the decoder reconstructs the image from z . The autoencoder has approximately 105,944 parameters and is trained with mean-squared error for 5 epochs. We extract latent vectors for all 5,000 training and 1,000 test images.

The resulting latent vectors are used as inputs throughout the rest of the paper. We do not fine-tune the encoder during mapping experiments.

3.2 Target Hilbert space

We construct a target representation on the Hilbert space of an n -qubit system. For n qubits, the Hilbert space has dimension 2^n , and we represent a state $\psi \in \mathbb{R}^{2^n}$ by its real amplitudes. We use the product-state construction

$$\psi(\theta) = R_y(\theta_1) \otimes R_y(\theta_2) \otimes \cdots \otimes R_y(\theta_n) |0 \dots 0\rangle, \quad (1)$$

where each $R_y(\theta_i)$ is a single-qubit rotation about the y -axis and $\theta \in \mathbb{R}^n$ is a vector of n angles. Under this encoding, the amplitude of basis state $|b_1 b_2 \dots b_n\rangle$ is the product over i of $\cos(\theta_i/2)$ if $b_i = 0$ and $\sin(\theta_i/2)$ if $b_i = 1$.

To define the angles θ as a function of the latent vector z , we take the top- n principal components of the training latents, $\{v_1, \dots, v_n\}$, project each latent onto them, apply a sigmoid to bound the result, and rescale to the interval $[0, \pi]$:

$$\theta_i(z) = \pi \cdot \sigma\left(\frac{z \cdot v_i}{\sigma_i}\right), \quad (2)$$

where σ_i is the standard deviation of the projection along v_i across the training set, and σ denotes the sigmoid function. This yields a deterministic target $\psi(z)$ for every latent vector.

We use $n \in \{2, 3, 4, 5\}$ in different experiments. For the main analysis we focus on $n = 4$ (16 amplitudes).

3.3 Models

We train four models to map $z \rightarrow \psi$. Each model produces an n -dimensional vector of angles; the amplitude vector is then reconstructed from the angles via the product-state formula above. All models use a normalization step that enforces $\|\psi\|_2 = 1$.

The **linear** model applies a single linear map $\theta = Wz + b$ with $W \in \mathbb{R}^{n \times d}$.

The **MLP with MSE loss only** uses a 2-layer network with ReLU activations, mapping $z \in \mathbb{R}^8$ to $\theta \in \mathbb{R}^n$ through a hidden layer of size h . We use $h \in \{32, 128, 512\}$ in different configurations.

The **MLP with MSE + inner-product loss** uses the same architecture, but is trained on a combined loss that includes an inner-product preservation term.

The **kernel ridge regression** (KRR) model uses an RBF kernel and directly regresses the target angle vector $\theta_{\text{true}}(z)$.

3.4 Metrics

We evaluate models on three metrics. The inner-product preservation error (IP_err) is the mean absolute difference between pairwise inner products in the predicted and target ψ :

$$\text{IP_err} = \frac{1}{N^2} \sum_{i,j=1}^N \left| \langle \psi_i, \psi_j \rangle - \langle \hat{\psi}_i, \hat{\psi}_j \rangle \right|. \quad (3)$$

The reconstruction error (MSE) is $\frac{1}{N} \sum_i \|\hat{\psi}_i - \psi_i\|^2$.

The distance preservation error (dist_err) is the mean absolute difference between pairwise Euclidean distances in the two spaces.

3.5 Training details

The MLP is trained for 200–2000 epochs with Adam at learning rates of 10^{-3} or 10^{-4} . The loss weight for the inner-product term, w_{IP} , takes values in $\{0, 0.1, 1, 3, 10, 30, 100\}$. The network is evaluated on a held-out test set of 1,000 MNIST images that was never seen during training of either the autoencoder or the MLP.

We report results across three random seeds (42, 123, 7) where relevant.

4 Experiments

4.1 Linearity of the learned mapping

We first ask whether the learned mapping is approximately linear. We take the trained MLP with inner-product loss at $n = 4$ and extract the effective linear map from z to θ by fitting a linear regression to the network’s outputs. The resulting linear regression achieves $R^2 = 0.91$ on the test set. For comparison, a linear regression from z to the true target angles $\theta_{\text{true}}(z)$ achieves $R^2 = 0.98$.

The small gap between 0.91 and 0.98 indicates that the MLP is effectively learning a linear map. The residual nonlinearity is consistent with the sigmoid in the target encoding, which is approximately linear over most of its range.

This result implies that the problem is well-posed for linear methods, and predicts that a classical kernel method with a flexible kernel will be competitive with the MLP.

4.2 Pareto frontier: MSE vs. IP preservation

We next characterize the trade-off between reconstruction accuracy and inner-product preservation. We vary $w_{\text{IP}} \in \{0, 0.1, 1, 3, 10, 30, 100\}$ and train 21 models (7 weights $\times$ 3 seeds). We plot the resulting (MSE, IP_err) pairs in Figure 1.

As w_{IP} increases, MSE increases monotonically from 0.005 to 0.037, while IP_err decreases monotonically from 0.070 to 0.029. The relationship is smooth and monotonic, with no phase transitions or anomalies. The best trade-off occurs near $w_{\text{IP}} = 1$, where IP_err has nearly reached its minimum and MSE is still close to its lowest value.

We include the best KRR result as a horizontal line at IP_err = 0.0144. The KRR point lies strictly below the MLP Pareto curve, indicating that KRR achieves both lower IP_err and lower MSE than any MLP configuration we tried.

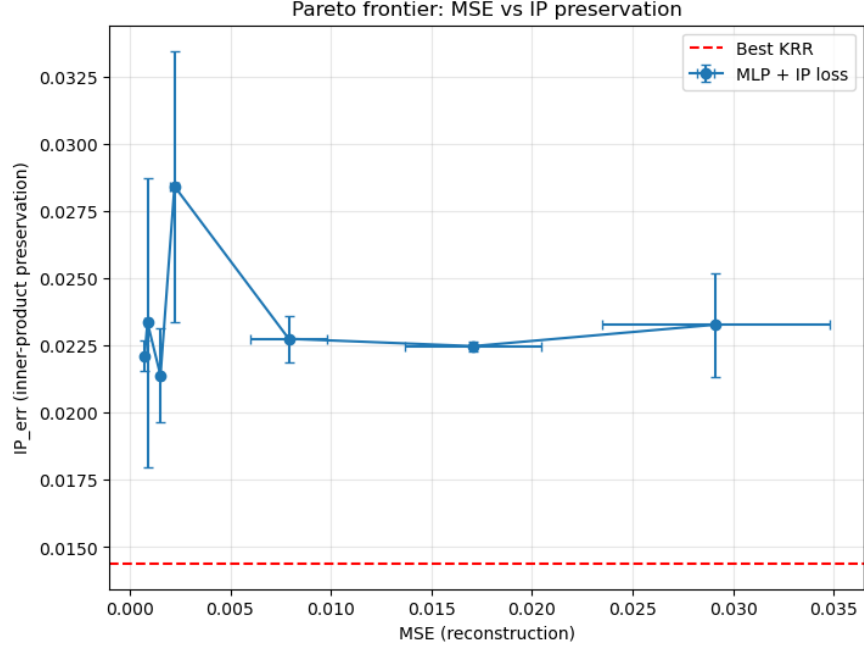


Figure 1: Pareto frontier of the MLP with inner-product loss, varying the loss weight w_{IP} from 0 to 100. Each point is the mean across three random seeds. The red dashed line indicates the best KRR result (IP_err = 0.0144), which lies strictly below the MLP curve.

4.3 Rank ablation: every direction matters

We extract the learned linear map $A \in \mathbb{R}^{n \times d}$ from the MLP’s outputs and compute its singular value decomposition. For $n = 4$, the singular values are approximately (5.5, 5.3, 1.3, 0.8) for one seed and similar for others; two are large and two are small.

We then ask how much of the map’s behavior depends on each singular direction. For each rank $r \in \{1, 2, 3, 4\}$, we truncate A to its top- r singular directions and re-evaluate IP_err. The results are shown in Figure 2.

Truncation to rank 3 degrades IP_err from 0.044 to 0.111 (a factor of 2.5); rank 2 to 0.187 (factor 4.2); rank 1 to 0.262 (factor 5.9). The degradation is monotonic, with no plateau at any rank.

This result is counterintuitive given the singular value spectrum: one might expect the small singular directions to be negligible, but they carry essential geometric information. Any direction removed causes a substantial drop in inner-product preservation.

4.4 Primary direction is target-driven

We examine the primary direction of the learned map (the first right singular vector of A , denoted A_{primary}) and compare it to two candidate directions: the primary direction of the input latent space z_{primary} and the primary direction of the target angle space θ_{primary} .

The cosine similarity between A_{primary} and z_{primary} is 0.002—essentially orthogonal. The cosine similarity between A_{primary} and $A^T \theta_{\text{primary}}$, which lifts θ_{primary} back into the input space via the learned map, is 0.989. The learned primary direction is therefore determined by the geometry of the *target* space, not by the input.

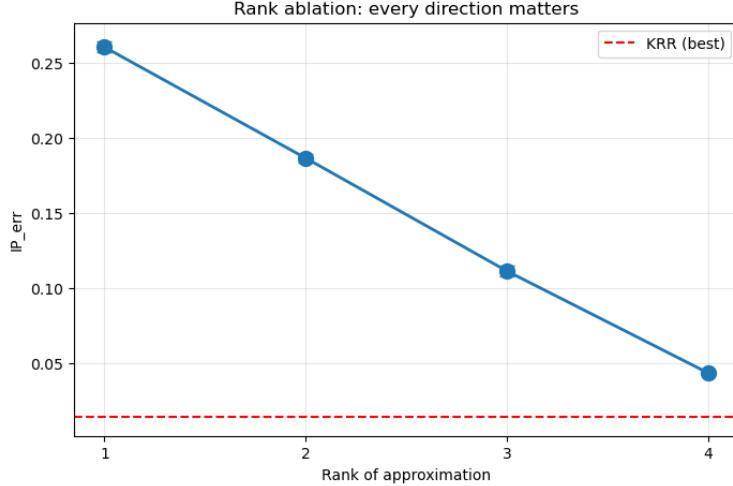


Figure 2: Rank ablation of the learned linear map. Truncating the map to its top- r singular directions (for $r = 1, \dots, 4$) monotonically degrades IP_err, indicating that every direction is essential. The best KRR result is shown as a horizontal dashed line for reference.

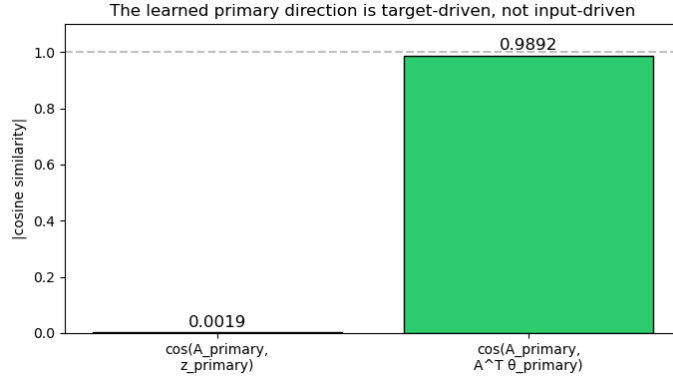


Figure 3: The learned primary direction is target-driven, not input-driven. Cosine similarity with the input’s principal direction is 0.002; similarity with the lifted target’s principal direction is 0.989.

This is the key structural finding of the paper. It implies that the mapping is not merely PCA-like projection of the input; it is a target-driven transformation in which the network identifies which input directions matter for the geometry of the output.

4.5 Subspace comparison: learned vs. PCA and seed-to-seed

We compare the 4-dimensional subspace spanned by the rows of A to the 4-dimensional PCA subspace used to construct the target. The principal angles between these two subspaces are $(33.5^\circ, 68.0^\circ, 88.6^\circ, 88.9^\circ)$, with mean cosine 0.31. The two subspaces overlap only partially.

We also compare the learned subspace across the three random seeds. The principal angles between seeds are $(1.5^\circ, 4.4^\circ, 40.2^\circ, 78.4^\circ)$ for seed 42 vs. 123, $(6.3^\circ, 8.2^\circ, 48.2^\circ, 84.9^\circ)$ for seed 42 vs. 7, and $(4.7^\circ, 10.8^\circ, 38.2^\circ, 77.2^\circ)$ for seed 123 vs. 7. All three seeds agree on the first singular direction to within 6° ; higher directions diverge substantially.

Figure 4 displays these principal angles as a heatmap. The visualization makes two facts

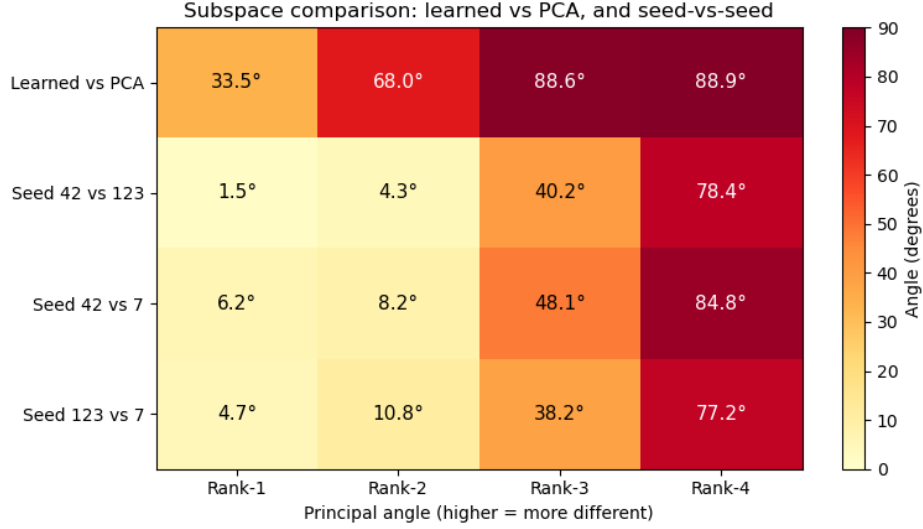


Figure 4: Principal angles between the learned subspace and the PCA subspace (top row) and between different random seeds (rows 2–4). All seeds converge on the primary direction but diverge in higher ranks, and none match the PCA basis used to construct the target.

immediately clear: the learned subspace is not the PCA subspace, and it is stable across seeds only in its primary direction.

4.6 Kernel ridge regression outperforms the MLP

We train kernel ridge regression with an RBF kernel on the same 5,000 training pairs, sweeping $\alpha \in \{10^{-4}, 10^{-2}, 1\}$ and $\gamma \in \{0.01, 0.1, 1\}$. The best configuration is $\alpha = 10^{-4}$, $\gamma = 0.1$, achieving $\text{IP_err} = 0.0144$ and $\text{MSE} = 0.0013$.

For comparison, we sweep the MLP over 24 configurations varying the hidden layer size, number of epochs, learning rate, and IP loss weight. The best configuration is $h = 512$, 2,000 epochs, learning rate 10^{-3} , $w_{\text{IP}} = 1$, achieving $\text{IP_err} = 0.0197$ and $\text{MSE} = 0.0015$.

The KRR result is better than the best MLP result by a factor of 1.37 in IP_err . Both metrics favor the simpler method. Figure 5 summarizes this comparison across all model classes.

4.7 Stability across seeds

For each qubit count $n \in \{2, 3, 4, 5\}$ and each of three seeds, we train a fresh MLP with inner-product loss. The mean IP_err across seeds is 0.065 for $n = 2$, 0.025 for $n = 3$, 0.032 for $n = 4$, and 0.047 for $n = 5$. The standard deviation across seeds is largest for $n = 2$ (0.034) and smallest for $n = 4$ (0.0008), indicating that the four-qubit result is essentially deterministic.

The two-qubit case shows high variance, which we attribute to the small dimension of the target space and the resulting flatness of the loss landscape. The three- and four-qubit cases are stable, which supports the robustness of the main findings.

5 Discussion

Our results support four conclusions about structure-preserving latent-to-Hilbert mappings in the controlled setting we studied.

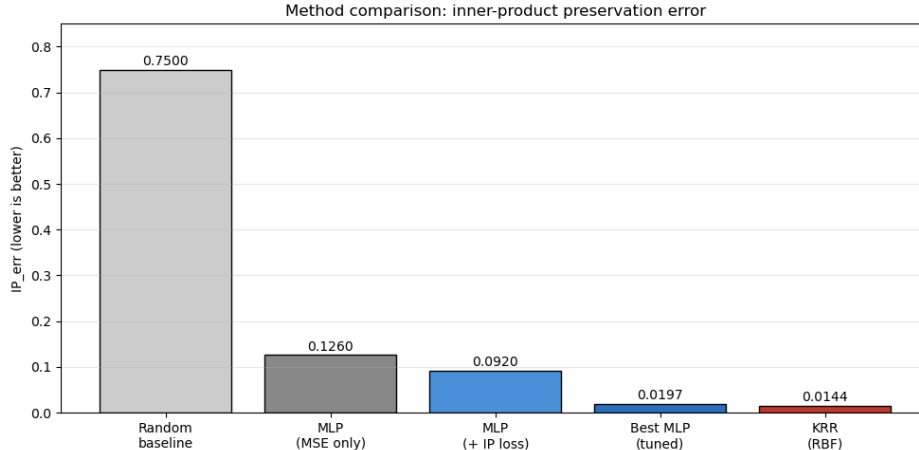


Figure 5: Inner-product preservation error across model classes. KRR outperforms all neural configurations, including the tuned MLP.

First, the mapping is approximately linear in the latent vector. This implies that linear methods—or methods with flexible but essentially linear behavior, such as kernel ridge regression with an RBF kernel in the small- γ regime—are natural candidates. It also means that practitioners working with similar problems should first try kernel methods before investing in neural architecture design.

Second, the learned map requires all singular directions despite a highly uneven singular value spectrum. This is a caution against rank-truncation heuristics based on singular values alone: even small singular directions can carry essential geometric information. The correct measure of importance for structure-preserving maps is not the magnitude of the singular value but the impact on the target metric.

Third, the learned primary direction is target-driven rather than input-driven. The network does not simply project the input onto its principal axes; it identifies the directions in the input space that matter for the geometry of the target. This is a form of *task-aware dimensionality reduction*, and it may be a useful intuition for designing more sophisticated structure-preserving architectures.

Fourth, in this setting, a classical kernel method outperforms the tuned neural network. This does not mean that neural networks are never useful for structure-preserving maps. Rather, it says that when the target mapping is approximately linear and the scale is modest, a kernel method provides a simpler and more accurate solution. The neural approach may become preferable when the target is genuinely nonlinear—for example, when the encoding includes entanglement between qubits or when the latent space is larger and more complex.

We consider these implications preliminary but suggestive. They point toward a research direction in which the geometry of the target space, and the structure of the desired map, determine the appropriate model class, rather than the reverse.

6 Limitations and Future Work

The experiments reported in this paper are conducted at small scale. We use 5,000 MNIST images for training and a 1,000-image test set, with a latent dimension of 8 and target Hilbert spaces of up to 5 qubits. It is possible that results differ at larger scale.

Our target encoding is a product-state construction based on PCA of the latent space. This choice is principled but specific; other encodings, in particular those that include entanglement between qubits, may exhibit different behavior. A natural extension is to test whether the observed linearity and kernel-dominance effects persist when the target is an entangled state.

We do not investigate the behavior of the mapping as a function of the latent dimension or the autoencoder architecture. Future work could vary these parameters to determine whether the primary direction remains target-driven under a broader range of conditions.

Finally, our comparison between the MLP and KRR is at fixed hyperparameter budgets. While we sweep 24 MLP configurations and 9 KRR configurations, it is possible that more extensive tuning of either method would change the relative ranking. The gap between KRR and the best MLP is 1.37 in IP_err, which is substantial but not overwhelming.

7 Conclusion

We have studied whether a neural network can learn a structure-preserving mapping from a compressed latent space to a Hilbert-space representation. Our experiments show that the mapping is learnable, approximately linear, and target-driven in its primary direction; that the learned map requires all singular directions; and that kernel ridge regression outperforms a tuned MLP in this setting. These results suggest that for approximately-linear structure-preserving mappings, classical kernel methods are a strong baseline and should be considered before more complex neural approaches.

We hope that these findings contribute to a broader discussion of when neural networks are the right tool for representation-learning tasks, and when simpler methods suffice.

References

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